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MINOR PROJECT REPORT ON

TRYVIRTUAL

A Web-Based Apparel Customization and 3D Visualization Platform

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Abstract

The rapid growth of e-commerce platforms has significantly increased the demand for personalized products, particularly in the apparel industry. Traditional online shopping platforms generally provide only static product previews, which often fail to accurately represent how customized designs will appear on actual garments. This limitation can reduce customer confidence and lead to dissatisfaction after purchase. To address this issue, the project “**TryVirtual**” has been developed as a web-based apparel customization and visualization platform that enables users to design and preview customized clothing interactively.

TryVirtual provides a real-time apparel customization system where users can modify product designs by adding text, images, and color variations using an intuitive canvas-based editor. The platform is designed with a responsive user interface that adapts seamlessly across multiple devices, including mobile phones, tablets, and desktop systems. The modular component-based architecture ensures maintainability and scalability while providing a clean and modern user experience.

The development of TryVirtual follows the **Agile Software Development Life Cycle (SDLC)** model, allowing iterative improvements and continuous feature integration throughout the development process. The system design incorporates UML diagrams, responsive design principles, and modular programming practices to ensure maintainability and scalability.

Overall, TryVirtual demonstrates an effective implementation of modern web technologies to create an interactive and user-friendly apparel customization platform. The project highlights the practical application of component-based architecture, responsive design, and modern web frameworks in developing scalable web-based solutions for the e-commerce industry.

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Chapter 1

Introduction

1.1 Introduction to the Project

In recent years, the rapid growth of the internet and digital technologies has significantly transformed the way businesses interact with customers. One of the most prominent sectors that has experienced this transformation is the e-commerce industry. Online platforms now allow users to browse, compare, and purchase products conveniently from anywhere in the world. However, despite the convenience offered by online shopping platforms, certain limitations still exist, particularly when it comes to purchasing customizable products such as apparel.

Traditional online shopping websites often display static images of clothing items, allowing customers only to select predefined designs or colors. This approach restricts the user's ability to visualize personalized designs on the product before purchasing it. As a result, customers may face uncertainty about how their customized design will actually appear when printed on the final product.

To address this limitation, the project **“TryVirtual – A Web-Based Apparel Customization and 3D Visualization Platform”** has been developed. The primary aim of this project is to provide an interactive platform that allows users to customize apparel designs in real time and visualize the final output digitally before making a purchase decision.

The TryVirtual platform enables users to design apparel products such as T-shirts using an intuitive canvas-based interface where they can add text, upload images, adjust colors, and position design elements according to their preferences. The platform is designed with a modern and responsive user interface, ensuring compatibility across multiple devices including smartphones, tablets, and desktop computers.

The system is built using modern web development technologies such as **Next.js**, **TypeScript**, and **Tailwind CSS**. These technologies allow the application to deliver a fast, scalable, and highly responsive user experience. The modular component-based

architecture ensures that different sections of the application, such as the header, customization canvas, toolbar, and sidebar, can be developed and maintained independently.

Additionally, the platform supports features such as saving customization drafts, integrating with e-commerce workflows, and providing real-time previews of customized designs. These features enhance the user experience and make the system suitable for integration into modern e-commerce platforms.

Overall, TryVirtual demonstrates how advanced web technologies and interactive design tools can be used to improve the online shopping experience for customizable products. By allowing users to visualize their designs before purchase, the platform reduces uncertainty and increases customer confidence.

1.2 Project Category

The project **TryVirtual** falls under the category of **Internet-Based Application Development**, specifically within the domain of **Web-Based Product Customization Systems**. Such systems are designed to enable users to interact with digital interfaces to personalize products according to their preferences.

Internet-based application development involves designing and implementing software applications that operate over the web and can be accessed using standard web browsers. These applications rely on web technologies such as HTML, CSS, JavaScript, and modern frameworks to deliver dynamic and interactive user experiences.

The TryVirtual platform is a web-based application that focuses on product customization and visualization. It combines several modern development practices, including component-based architecture, responsive design, and modular development. The application runs entirely within a web browser, eliminating the need for users to install additional software or applications.

The system is also designed to integrate easily with existing e-commerce platforms. By functioning as a modular customization component, TryVirtual can be embedded into online stores to allow customers to personalize products before purchasing them.

This project also incorporates concepts from fields such as user interface design, web graphics rendering, and front-end software engineering. Therefore, it represents a multidisciplinary approach to solving real-world problems in the e-commerce domain.

1.3 Objectives of the Project

The main objective of the TryVirtual platform is to develop a modern and user-friendly web-based system that allows users to customize apparel products interactively. The project focuses on improving the overall user experience by enabling customers to design products visually before placing an order.

The primary objectives of the project are as follows:

- To design and develop an interactive web-based apparel customization platform.
- To implement a canvas-based editing environment that allows users to add text, images, and color modifications to apparel designs.
- To provide a responsive and intuitive user interface that works across different devices including mobile phones, tablets, and desktop computers.
- To create a modular and scalable software architecture using modern web technologies such as Next.js and TypeScript.
- To enable integration with existing e-commerce systems for product selection and cart management.
- To provide functionality for saving customization drafts so that users can continue editing their designs later.
- To ensure a visually appealing and user-friendly design using modern UI frameworks such as Tailwind CSS and Lucide React icons.
- To demonstrate the application of component-based architecture and modern software development practices in real-world web applications.

1.4 Problem Formulation

The growing demand for personalized products in the apparel industry has created a need for efficient digital customization platforms. Many existing e-commerce websites offer limited customization features, often restricted to selecting predefined colors or simple text additions. These limitations prevent users from fully expressing their creativity when designing customized apparel.

Another major issue with existing systems is the lack of interactive design tools. Users are often required to rely on static previews that do not accurately represent how the final product will look after printing. This lack of visualization can lead to dissatisfaction when the delivered product does not meet the user's expectations.

Furthermore, many existing customization tools are complex and difficult to use, especially for users who do not have prior experience with graphic design software. This creates a barrier for users who simply want to create personalized designs quickly and easily.

Therefore, the problem addressed by this project is the development of a user-friendly and interactive web-based system that allows users to design customized apparel products easily while providing accurate previews of their designs before purchase.

1.5 Identification and Recognition of Need

The need for a platform like TryVirtual arises from the increasing demand for personalization in modern consumer markets. Customers today prefer products that reflect their personal style and individuality. This trend is particularly evident in the fashion and apparel industry, where customized clothing has become highly popular.

However, traditional online shopping platforms do not provide sufficient tools for customers to create personalized designs. Users often have to rely on third-party design software or external services to create their designs before uploading them to the product customization system.

A dedicated platform that integrates design tools directly within the shopping experience can significantly improve usability and efficiency. By allowing users to create and preview their designs within the same platform, the customization process becomes more seamless and engaging.

Additionally, businesses benefit from such systems because they can offer personalized products without requiring manual design processing. Automated customization systems reduce operational costs while improving customer satisfaction.

Thus, the development of the TryVirtual platform addresses both user needs and business requirements by providing a modern solution for apparel customization.

1.6 Existing System

Currently, most online apparel platforms provide only basic product visualization and limited customization capabilities. These systems generally allow users to browse clothing items, select sizes, choose colors, and place orders. Some platforms provide simple image-based previews where users can see the product in different color variants. However, these systems are primarily based on static product images and do not provide interactive or immersive visualization features.

In the existing systems, customization features are often minimal and restricted to predefined templates. Users may be able to add simple text or select preset graphics, but the level of interaction remains limited. Additionally, many platforms lack advanced editing tools such as drag-and-drop positioning, real-time editing, or dynamic previews.

One of the major limitations of the existing systems is the absence of advanced visualization technologies. Most current platforms do not allow users to view apparel designs in a three-dimensional environment. As a result, customers cannot accurately understand how the customized design will appear on the product in real-world conditions.

The following limitations are commonly observed in existing systems:

- **Lack of 3D Visualization:** Most existing platforms only display static 2D product images. Users cannot visualize the apparel on a 3D model or rotate the product to see the design from multiple angles.
- **Limited Customization Tools:** Many systems offer only basic customization options such as selecting colors or adding predefined graphics.
- **No Real-Time Preview:** In many platforms, users cannot instantly see the result of their customization actions.
- **Poor User Interaction:** The absence of interactive editing tools such as drag-and-drop, resizing, or positioning reduces user engagement.
- **Lack of Draft Saving Features:** Some platforms do not allow users to save unfinished designs and continue later.
- **Limited Integration with Modern Web Technologies:** Traditional systems often use outdated architectures and lack support for modern frameworks that enable scalable and interactive applications.

Due to these limitations, users often face difficulty in accurately visualizing their customized apparel before placing an order. This creates uncertainty and may reduce customer satisfaction.

1.7 Proposed System

To overcome the limitations of the existing systems, the proposed system titled “**TryVirtual**” is designed as a modern web-based apparel customization and visualization platform. The system focuses on providing an interactive and user-friendly environment where users can design, customize, and visualize apparel products before making a purchase decision.

The proposed system integrates modern web technologies and design principles to deliver an efficient and scalable solution. The application is developed using the Next.js framework with TypeScript, which provides high performance, modular architecture, and server-side rendering capabilities. The system also utilizes Tailwind CSS for responsive and visually appealing user interfaces.

One of the major innovations of the proposed system is the integration of **3D visualization technology**. Unlike traditional platforms that display only static images, this system enables users to preview customized apparel designs on a three-dimensional model.

Users can rotate, zoom, and inspect the apparel from multiple angles, which provides a more realistic understanding of how the design will appear on the actual product.

The system also includes a canvas-based customization environment where users can modify different aspects of the apparel. They can add text, upload images, adjust colors, resize elements, and reposition design components in real time. This interactive editing process improves creativity and user engagement.

Additionally, the system is designed with a modular component-based architecture. Different interface elements such as the header, customization canvas, toolbar, sidebar, and preview panel are implemented as reusable components. This design approach improves maintainability, scalability, and code reusability.

The proposed platform is also capable of integration with existing e-commerce systems. It can connect with product databases, shopping cart services, and order management systems through REST APIs. This makes the system suitable for practical deployment in online clothing stores.

The key capabilities of the proposed system include:

- **Interactive 3D Visualization:** Users can view customized apparel designs on a 3D model, rotate the product, and inspect it from different viewing angles.
- **Real-Time Customization:** The platform provides a canvas-based editor that allows users to add text, upload graphics, and modify design elements dynamically.
- **Responsive User Interface:** The system is designed with a mobile-first approach and works seamlessly across desktops, tablets, and smartphones.
- **Modern Web Architecture:** Built using Next.js and TypeScript with a component-based structure for better scalability and maintainability.
- **Draft Saving Feature:** Users can save their unfinished designs and continue editing them later.
- **Integration with E-Commerce Platforms:** The system can connect with product catalogs, cart systems, and checkout processes through APIs.

By integrating advanced visualization techniques and modern web technologies, the proposed system provides a significantly improved user experience compared to existing apparel customization platforms. The ability to preview designs in a realistic 3D environment helps users make more informed decisions and increases confidence before purchasing customized apparel.

Overall, the TryVirtual system aims to bridge the gap between product customization and immersive visualization, making online apparel design more interactive, accurate, and engaging.

1.8 Unique Features of the System

The proposed system **TryVirtual** introduces several innovative and advanced features that distinguish it from traditional apparel customization platforms. These features focus on improving user interaction, enhancing visualization capabilities, and providing a modern and efficient design environment. The unique aspects of the system are described below.

1.8.1 3D Apparel Visualization

One of the most important and innovative features of the proposed system is the integration of **3D visualization technology**. In most existing customization platforms, users can only preview their design on a flat 2D image of the product. This approach often fails to provide an accurate understanding of how the design will appear on the real garment.

The TryVirtual platform addresses this limitation by allowing users to preview their customized design directly on a **three-dimensional apparel model**. The system maps the final canvas design onto a 3D T-shirt model and displays it within an interactive viewer. Users can rotate the model, zoom in or out, and inspect the apparel from different viewing angles.

This feature significantly improves the realism of the preview and allows users to better evaluate the final appearance of the product before purchasing it.

1.8.2 Real-Time Design Customization

The system provides a powerful **canvas-based customization editor** that enables users to modify apparel designs in real time. The editor allows users to add text elements, upload images, adjust colors, resize design components, and reposition objects on the canvas.

The real-time editing capability ensures that users can instantly view the effects of their modifications without refreshing the page or reloading the application. This interactive design experience improves usability and encourages creativity among users.

The customization editor is implemented using a canvas rendering library that efficiently handles graphical elements and supports advanced manipulation features such as scaling, rotation, and layering.

1.8.3 Modern Component-Based Architecture

Another important feature of the system is its **component-based architecture**. The entire user interface is divided into modular components such as header, toolbar, sidebar, design canvas, and preview panel.

Each component is designed to perform a specific function and can be reused throughout the application. This modular approach offers several advantages, including improved maintainability, easier debugging, and better scalability for future development.

Additionally, this architecture allows developers to extend the application by adding new components or modifying existing ones without affecting the entire system.

1.8.4 Responsive and Mobile-Friendly Interface

The proposed system follows a **responsive web design approach**, ensuring that the application works efficiently on different devices such as desktops, tablets, and smartphones.

The user interface automatically adapts to different screen sizes and resolutions, providing a consistent experience across devices. This is particularly important for e-commerce applications where users may access the platform from mobile devices.

Responsive design improves accessibility and allows users to customize products conveniently from any device.

1.8.5 Draft Saving and Design Continuation

The platform includes a **design draft saving feature** that allows users to save their unfinished customization work. This functionality ensures that users do not lose their progress and can continue editing their designs at a later time.

This feature is especially useful for users who want to experiment with multiple design ideas before finalizing their product. Saved drafts can be retrieved easily and modified further whenever required.

1.8.6 Integration with Existing E-Commerce Platforms

The TryVirtual system is designed to integrate seamlessly with existing e-commerce infrastructures. The platform can communicate with external systems using REST APIs to retrieve product information, manage shopping carts, and process customer orders.

This integration capability allows the customization module to be embedded within existing online stores without requiring a complete redesign of the e-commerce platform.

1.8.7 User-Friendly and Modern Interface Design

The system emphasizes a clean and modern interface design that enhances usability and visual appeal. The interface incorporates modern UI elements such as smooth transitions, hover effects, shadows, and interactive buttons.

A well-organized layout ensures that users can easily access different customization tools without confusion. The intuitive design reduces the learning curve for new users and provides a pleasant user experience.

1.8.8 High Performance Web Application

The system is built using modern web technologies that ensure high performance and efficiency. The application leverages optimized rendering techniques and efficient state management to maintain smooth interactions even when handling complex design elements.

Fast loading times, responsive interactions, and optimized rendering make the platform suitable for real-time customization tasks.

Overall, the unique features of the TryVirtual system combine modern web technologies with advanced visualization capabilities to create a powerful apparel customization platform. The integration of 3D visualization, real-time editing tools, responsive design, and modular architecture significantly enhances the user experience and addresses the limitations of existing customization systems.

Chapter 2

Requirement Analysis and System Specification

2.1 Introduction

Requirement analysis is one of the most critical phases in the software development life cycle (SDLC). It involves identifying, understanding, and documenting the functional and non-functional requirements of the system before the development process begins. Proper requirement analysis helps developers understand the objectives of the system, the expectations of users, and the limitations of the environment in which the system will operate.

For the **TryVirtual** platform, requirement analysis focuses on identifying the needs of users who want to customize apparel products online and preview them before purchasing. The system must provide an interactive design environment, support multiple customization options, and deliver a realistic visualization of the final product.

The analysis phase also involves determining the technical feasibility of implementing advanced features such as canvas-based design editing and three-dimensional apparel visualization. Since the platform is intended to integrate with existing e-commerce systems, it must also support compatibility with modern web technologies and APIs.

This chapter presents a detailed study of the system requirements, feasibility considerations, and specifications necessary for the successful development of the TryVirtual platform.

2.2 Feasibility Study

A feasibility study is conducted to determine whether the proposed system is practical and achievable within the given constraints of time, resources, and technology. It helps evaluate the viability of the project before significant resources are invested in development.

The feasibility study for the TryVirtual platform focuses on three major aspects: **technical feasibility, economic feasibility, and operational feasibility.**

2.2.1 Technical Feasibility

Technical feasibility evaluates whether the required technology, tools, and infrastructure are available to develop and implement the proposed system.

The TryVirtual platform is built using modern web development technologies such as **Next.js, TypeScript, Tailwind CSS, and React-based component architecture.** These technologies are widely used in modern web applications and provide strong support for building scalable and high-performance systems.

For the implementation of the design editor, canvas-based libraries can be used to manipulate graphical elements efficiently. These libraries allow users to add images, text, and shapes, and modify them dynamically on a design canvas.

The 3D visualization feature is implemented using **Three.js**, a powerful JavaScript library used for rendering interactive 3D graphics within web browsers. Three.js allows the system to load 3D apparel models and apply textures representing the customized design created by the user.

Since all the required technologies are open-source and well-documented, the development process becomes more efficient and manageable. Additionally, modern browsers support WebGL, which enables smooth rendering of 3D graphics directly within the browser.

Therefore, from a technical perspective, the development of the TryVirtual platform is feasible and achievable using available technologies.

2.2.2 Economic Feasibility

Economic feasibility evaluates whether the benefits of the proposed system justify the costs required for development and implementation.

The TryVirtual platform is designed primarily using open-source tools and frameworks such as Next.js, Tailwind CSS, and Three.js. These technologies do not require licensing fees, which significantly reduces the development cost.

The main expenses involved in the project include development time, hosting services, and maintenance costs. However, the system can be deployed on cloud-based platforms

such as Vercel, which offer affordable and scalable hosting solutions.

From a business perspective, the platform can increase customer engagement and improve online shopping experiences by allowing users to visualize customized apparel before purchasing. This can lead to increased sales, reduced product returns, and higher customer satisfaction.

Considering these factors, the economic investment required for the development of the TryVirtual system is relatively low compared to the potential benefits it offers.

2.2.3 Operational Feasibility

Operational feasibility determines whether the proposed system can be effectively used and maintained within the intended operational environment.

The TryVirtual platform is designed with a user-friendly interface that allows customers to easily customize apparel designs without requiring technical knowledge. The system provides intuitive tools for adding text, uploading images, adjusting colors, and positioning design elements on the canvas.

The integration of the 3D visualization feature further enhances the usability of the platform by allowing users to preview their designs in a realistic environment. This helps customers make better purchasing decisions and improves overall user satisfaction.

From the perspective of online retailers, the system can be easily integrated into existing e-commerce websites. Retailers can provide customers with a “Customize” option that opens the TryVirtual interface, allowing them to design and preview their apparel products before adding them to the shopping cart.

The system also supports modular development, which simplifies future updates and maintenance. Developers can easily add new features or modify existing components without affecting the entire system.

Therefore, the TryVirtual platform is operationally feasible and suitable for deployment in modern e-commerce environments.

2.3 Software Requirement Specification (SRS)

The Software Requirement Specification (SRS) describes the functional and non-functional requirements of the TryVirtual platform. It defines the expected behavior of the system and provides a clear understanding of how the application should operate.

The requirements are categorized into several important aspects including data requirements, functional requirements, performance requirements, dependability, maintainability, security, and user interface design.

2.3.1 Data Requirements

Data requirements define the type of data that will be stored, processed, and managed by the system. The TryVirtual platform handles different categories of data related to user customization and product information.

- User design data including uploaded images, text elements, colors, and positioning information.
- Product information such as apparel type, size, available colors, and product identifiers.
- Canvas design configuration including object layers, transformations, and layout settings.
- Draft design data allowing users to save incomplete designs for future editing.
- Product preview data used to generate the final texture for the 3D apparel model.
- Shopping cart integration data used for transferring customized product information to the e-commerce checkout system.

2.3.2 Functional Requirements

Functional requirements describe the core operations and features that the system must perform.

- The system should allow users to open the customization interface from an e-commerce product page.
- The system should provide a canvas-based design editor for apparel customization.
- Users should be able to upload images and place them on the apparel design area.
- Users should be able to add text elements and modify their font style, size, and color.
- The system should allow resizing, rotating, and repositioning of design elements.
- The platform should generate a real-time preview of the customized design on the canvas.
- The system should allow users to save their design as a draft.
- The system should allow users to load previously saved drafts and continue editing.

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- The system should provide a 3D preview feature that maps the final design onto a 3D apparel model.
 - Users should be able to rotate and zoom the 3D model for better visualization.
 - The system should allow the finalized design to be sent to the shopping cart for purchase.

2.3.3 Performance Requirements

Performance requirements define how efficiently the system should operate under different conditions.

- The system should load the customization interface within a few seconds.
- Canvas editing operations such as resizing and moving elements should occur in real time.
- The system should render the 3D model smoothly without noticeable lag.
- The application should support multiple users accessing the platform simultaneously.
- The platform should maintain responsive performance across different devices and browsers.

2.3.4 Dependability Requirements

Dependability ensures that the system operates reliably and consistently.

- The system should prevent accidental loss of user design data.
- The platform should handle errors gracefully and provide appropriate messages to users.
- The system should ensure that uploaded files are processed correctly.
- The system should maintain consistent design previews across sessions.

2.3.5 Maintainability Requirements

Maintainability requirements define how easily the system can be updated and maintained.

- The application should follow a modular component-based architecture.

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- The codebase should follow standardized coding practices.
 - The system should allow developers to add new customization tools without affecting existing components.
 - Documentation should be maintained for easier system updates and debugging.

2.3.6 Security Requirements

Security requirements ensure that the system protects user data and prevents misuse.

- The system should validate uploaded files to prevent malicious content.
- User data and customization designs should be stored securely.
- API communication with e-commerce platforms should be protected using secure protocols.
- Unauthorized access to stored design drafts should be prevented.

2.3.7 Look and Feel Requirements

The user interface design plays an important role in improving the usability and visual appeal of the platform.

- The interface should have a clean and modern design.
- The layout should be intuitive and easy to navigate.
- The platform should use responsive design principles to support multiple screen sizes.
- Interactive elements such as buttons and controls should provide visual feedback.
- The overall user experience should be smooth and visually appealing.

2.4 Validation

Validation ensures that the system requirements and design decisions correctly address the needs of the users and the objectives of the project. It confirms that the proposed solution solves the identified problem effectively and performs as expected.

The validation process for the TryVirtual platform includes the following considerations:

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- Ensuring that the system requirements clearly reflect the needs of users who want to customize apparel online.
 - Verifying that the design editor provides all necessary customization tools such as text addition, image upload, resizing, and positioning.
 - Confirming that the system correctly maps the final design from the canvas onto the 3D apparel model.
 - Checking that the platform functions properly across different web browsers and devices.
 - Ensuring that the system integrates smoothly with existing e-commerce platforms for product purchasing.
 - Validating that the user interface is intuitive and easy to understand for first-time users.
 - Ensuring that the saved draft feature allows users to store and reload their designs successfully.

Proper validation helps guarantee that the TryVirtual system meets both technical and user expectations before deployment.

2.5 Expected Hurdles

During the development of a software project, several challenges and obstacles may arise. Identifying these hurdles in advance helps developers prepare strategies to address them effectively.

Some of the expected challenges in the development of the TryVirtual platform include:

- **3D Model Integration:** Mapping the 2D canvas design onto a 3D apparel model requires proper texture mapping and coordinate alignment.
- **Performance Optimization:** Rendering 3D models within a web browser may affect performance if not optimized properly.
- **Browser Compatibility:** Different web browsers may handle WebGL and graphical rendering differently.
- **Responsive Design Complexity:** Maintaining a consistent user experience across mobile devices, tablets, and desktops requires careful interface design.

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- **File Upload Management:** Handling user-uploaded images requires proper validation to avoid large file sizes or unsupported formats.
 - **State Management:** Managing real-time design data and user interactions efficiently can be complex in large applications.
 - **Integration with E-Commerce Platforms:** Connecting the customization system with existing shopping cart systems requires well-structured APIs.

Addressing these hurdles through proper design and testing will ensure the reliability and effectiveness of the TryVirtual system.

2.6 SDLC Model Used

The development of the TryVirtual platform follows the **Agile Software Development Life Cycle (SDLC) model**. Agile is an iterative and flexible development approach that allows teams to build software in small increments while continuously improving the system based on feedback.

The Agile model was selected for this project because it supports continuous development, rapid testing, and easy modification of features during the development process.

2.6.1 Reasons for Choosing Agile Model

The Agile methodology is suitable for the development of modern web applications due to the following reasons:

- It allows incremental development of features such as the design editor, customization tools, and 3D preview system.
- It supports frequent testing and debugging during development.
- It allows developers to adapt to changing requirements or design improvements.
- It improves collaboration between development stages.
- It ensures faster delivery of functional modules.

2.6.2 Stages of Agile Development Used in the Project

The TryVirtual system follows multiple iterative stages during development:

- **Requirement Analysis:** Identifying the needs of users who want to customize apparel products online.

-
- **System Design:** Designing the architecture of the platform including UI components, canvas editor, and 3D visualization module.
 - **Development:** Implementing the features using Next.js, TypeScript, Tailwind CSS, and other modern technologies.
 - **Testing:** Verifying that each feature works correctly and fixing bugs during development iterations.
 - **Deployment:** Deploying the web application on cloud platforms such as Vercel.
 - **Maintenance and Improvements:** Updating the system based on user feedback and adding new features.

The Agile model enables the project to evolve continuously, ensuring that the final system is efficient, scalable, and aligned with user expectations.

Chapter 3

System Design

3.1 Introduction

System design is the phase in the software development life cycle where the conceptual model of the system is transformed into a detailed architecture and design structure. It defines how different components of the system interact with each other and how the requirements identified in the previous phase are implemented in the software system.

The purpose of system design is to provide a clear blueprint for the development process. It describes the internal structure of the system, the data flow between different modules, the database organization, and the user interface design.

For the **TryVirtual** platform, the system design focuses on creating an efficient architecture that supports apparel customization, canvas-based editing, and 3D visualization of the final design. The system is designed using modern web technologies and follows a modular component-based architecture to ensure scalability, maintainability, and performance.

The design phase includes several important aspects of the system, such as:

- Designing the architecture of the web application.
- Defining how different modules interact with each other.
- Creating diagrams to represent the system structure and data flow.
- Designing the user interface for customization and visualization.
- Defining the structure of the database used for storing user data and designs.

3.2 Design Approach

The TryVirtual platform follows an **Object-Oriented Design (OOD)** approach combined with a component-based architecture. This design methodology helps in organizing

the system into modular and reusable components.

Object-oriented design focuses on representing system elements as objects that contain both data and functions. Each component of the application performs a specific task and interacts with other components through well-defined interfaces.

The major advantages of using an object-oriented design approach include:

- Improved modularity and code organization.
- Reusability of components across different parts of the application.
- Easier maintenance and debugging.
- Scalability for adding new features in the future.
- Better representation of real-world entities within the system.

In the TryVirtual system, the application is divided into multiple modules such as:

- **Header Module** – Provides navigation options such as returning to the store, viewing saved designs, and adding the customized product to the cart.
- **Canvas Editor Module** – Allows users to design and customize apparel by adding text, images, and other graphical elements.
- **Toolbar Module** – Contains editing tools such as resize, rotate, delete, and color adjustment.
- **Sidebar Module** – Provides options for uploading images, selecting fonts, and choosing design elements.
- **3D Preview Module** – Displays the final customized design on a 3D apparel model.
- **State Management Module** – Manages real-time updates of the design elements and user interactions.

These modules work together to provide a smooth and interactive customization experience for users.

3.3 Detailed System Design

The detailed design of the TryVirtual platform describes how each component of the system is implemented and how data flows between different modules.

The system architecture is divided into the following layers:

- **Presentation Layer**

- Responsible for displaying the user interface.
- Handles user interactions such as clicking buttons, editing designs, and uploading images.
- Implemented using Next.js, React components, and Tailwind CSS.

- **Application Logic Layer**

- Manages the business logic of the system.
- Handles design editing operations such as resizing, rotating, and repositioning objects.
- Processes data from the canvas editor and prepares it for preview and storage.

- **Visualization Layer**

- Responsible for rendering the 3D apparel model.
- Uses Three.js to display the model and apply the design texture.
- Allows users to rotate, zoom, and inspect the apparel model.

- **Integration Layer**

- Connects the customization platform with external e-commerce systems.
- Transfers product information and customized design data to the shopping cart.
- Handles communication through REST APIs.

This layered architecture ensures that the system remains organized and scalable while maintaining clear separation between different functional components.

3.4 Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a graphical representation used to illustrate how data moves through a system. It shows the flow of information between different processes, data stores, and external entities.

DFDs help in understanding the logical flow of the system and identifying how data is processed at different stages.

For the TryVirtual platform, the data flow diagrams describe how user inputs such as uploaded images, text elements, and design settings are processed by the system and eventually displayed as a customized apparel preview.

3.4.1 DFD Level 0 (Context Diagram)

The Level 0 Data Flow Diagram, also known as the context diagram, represents the overall system as a single process and shows how it interacts with external entities.

The main external entities involved in the TryVirtual system include:

- **User** – Interacts with the system to design and customize apparel.
- **E-Commerce Platform** – Receives the customized product information when the user adds the product to the cart.
- **Database/Storage System** – Stores user design drafts and product information.

The Level 0 DFD shows the general flow of data between the user, the TryVirtual system, and the external systems.

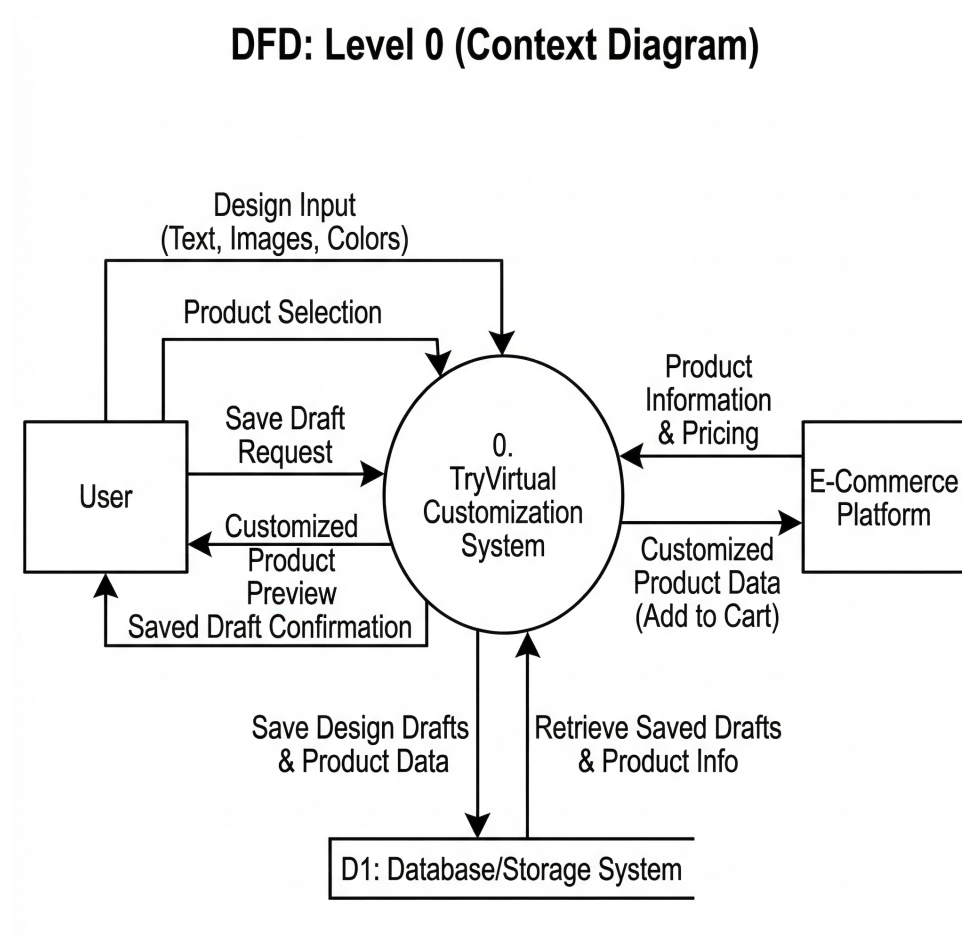


Figure 3.1: DFD Level 0 (Context Diagram) of TryVirtual System

3.4.2 DFD Level 1

The Level 1 Data Flow Diagram provides a more detailed view of the internal processes within the TryVirtual system. It breaks the system into multiple processes that handle

different functionalities.

Major processes involved in the Level 1 DFD include:

- **Design Editor Process** – Handles canvas editing operations such as adding text, uploading images, and modifying design elements.
- **Design Processing Module** – Converts the canvas design into a format suitable for preview and storage.
- **3D Visualization Module** – Applies the final design as a texture to the 3D apparel model and displays it to the user.
- **Draft Management Module** – Allows users to save and load design drafts.
- **Cart Integration Module** – Sends customized product data to the e-commerce shopping cart.

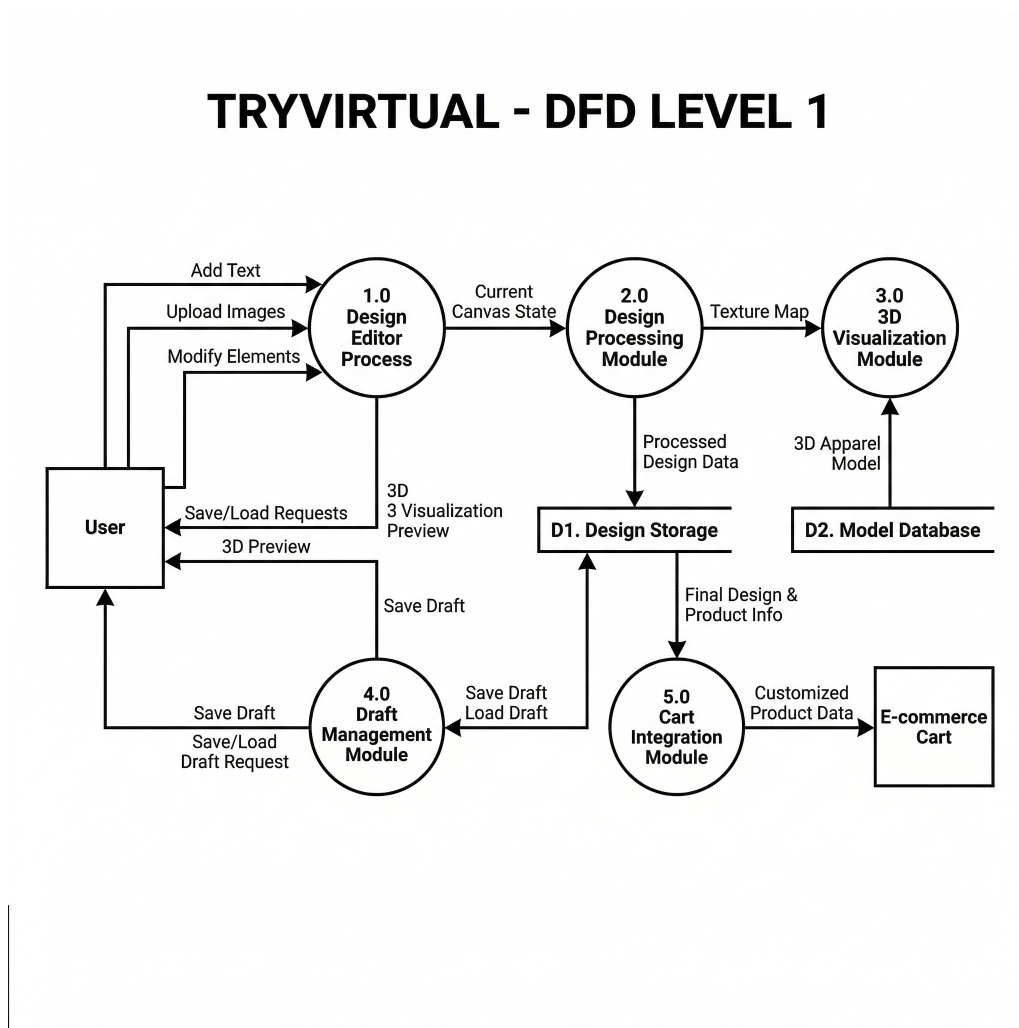


Figure 3.2: DFD Level 1 of TryVirtual System

3.5 System Flowchart

A flowchart is a diagram that represents the step-by-step flow of operations in a system. It helps visualize the logical sequence of processes involved in completing a task.

The flowchart of the TryVirtual platform illustrates how a user interacts with the customization system from start to finish.

The general workflow of the system includes the following steps:

- User opens the product customization page.
- The system loads the apparel customization interface.
- The user adds text or uploads images to the design canvas.
- The user edits and adjusts design elements.
- The system generates a real-time preview of the design.
- The user can choose to save the design as a draft.
- The user can view the design on a 3D apparel model.
- After finalizing the design, the user adds the customized product to the shopping cart.

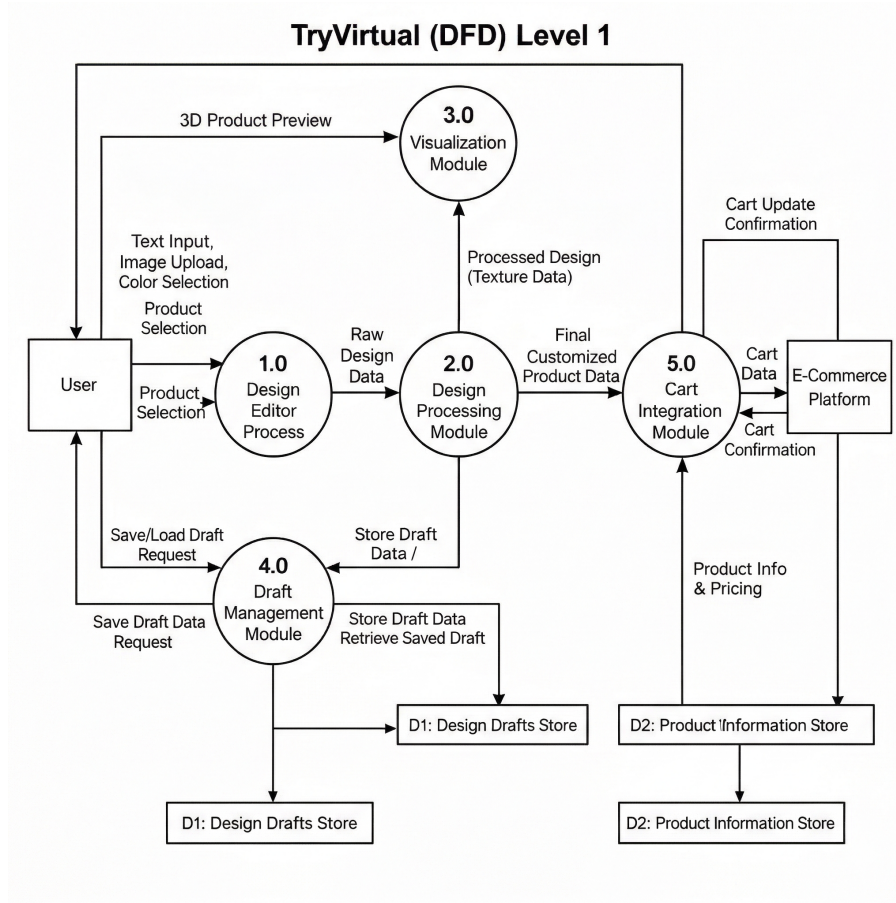


Figure 3.3: System Flowchart of TryVirtual Platform

3.6 Data Dictionary

A data dictionary provides detailed information about the data elements used within the system. It helps developers understand how data is structured and stored.

The key data elements used in the TryVirtual system include:

- **UserID** – Unique identifier for each user.
- **DesignID** – Unique identifier for each saved design.
- **ImageData** – Stores uploaded image files used in the design.
- **TextElement** – Stores text content added to the design.
- **FontStyle** – Stores the font type used for text elements.
- **ColorValue** – Represents the color applied to design elements.
- **CanvasLayout** – Stores the position, scale, and rotation of design objects.
- **3DTexture** – Represents the final texture generated from the canvas design.

The data dictionary ensures consistency in how data is used throughout the system.

3.7 Structured Chart

A structured chart is used to represent the hierarchical relationship between different modules of the system. It helps illustrate how different functions of the system are organized and connected.

The main modules in the TryVirtual system include:

- **User Interface Module**
 - Header Component
 - Navigation Controls
 - Customization Interface
- **Design Editor Module**
 - Canvas Rendering
 - Text Editing Tools
 - Image Upload Handler
 - Object Transformation Tools
- **Preview Module**
 - 2D Design Preview
 - 3D Model Visualization
 - Model Rotation and Zoom Controls
- **Data Management Module**
 - Draft Saving System
 - Design Retrieval
 - Data Storage Handling
- **Integration Module**
 - Product Data API
 - Cart Integration
 - Order Processing Connection

3.8 UML Diagrams

Unified Modeling Language (UML) diagrams are used to visually represent the structure and behavior of a software system. These diagrams help developers and stakeholders understand how different components of the system interact with each other.

For the TryVirtual platform, several UML diagrams are used to represent the functionality and architecture of the system. These diagrams provide a clear visualization of system operations, user interactions, and relationships between different components.

The UML diagrams used in this project include:

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram

3.8.1 Use Case Diagram

A Use Case Diagram represents the interaction between users and the system. It identifies the main functionalities that the system provides and the actors who interact with those functionalities.

In the TryVirtual system, the primary actor is the user who wants to customize apparel products online.

The main use cases of the system include:

- Opening the apparel customization interface.
- Uploading images or graphics for customization.
- Adding and editing text elements on the apparel.
- Adjusting design elements such as size, rotation, and position.
- Saving the design as a draft.
- Previewing the design on a 3D apparel model.
- Adding the customized product to the shopping cart.

The use case diagram illustrates the relationship between the user and these system functionalities.

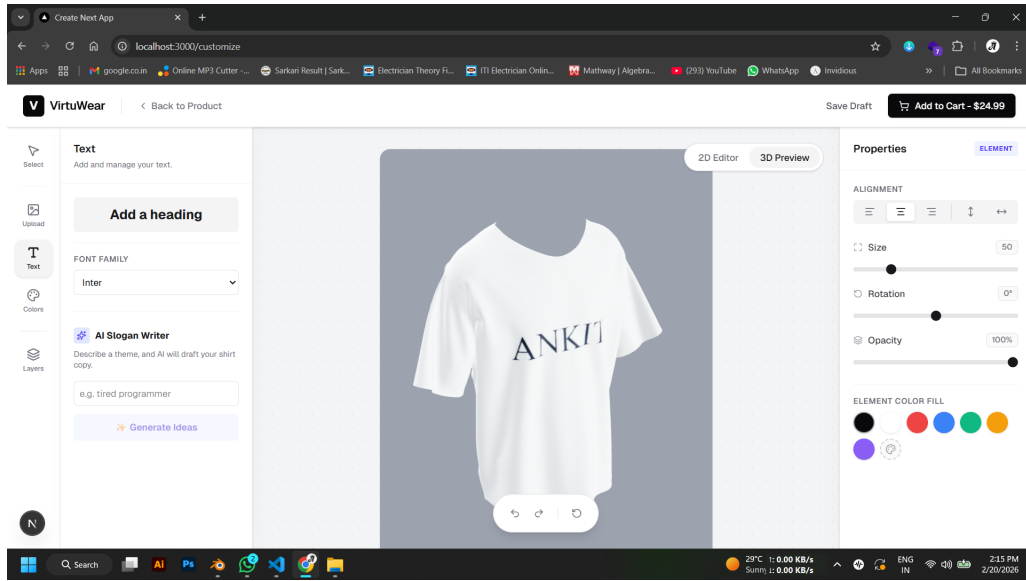


Figure 3.4: Use Case Diagram of TryVirtual System

3.8.2 Class Diagram

A Class Diagram represents the structure of the system by showing the classes, their attributes, methods, and relationships with other classes.

The TryVirtual system uses a component-based architecture where different components interact with each other to provide the customization functionality.

Important classes in the system include:

- **User Class**

- Attributes: UserID, Username, Email
- Methods: login(), saveDesign(), loadDesign()

- **Design Class**

- Attributes: DesignID, CanvasData, CreatedDate
- Methods: createDesign(), editDesign(), deleteDesign()

- **CanvasEditor Class**

- Attributes: canvasObjects, objectProperties
- Methods: addText(), uploadImage(), resizeObject(), rotateObject()

- **Preview Class**

- Attributes: modelTexture, previewState

-
- Methods: generatePreview(), render3DModel()

- **CartIntegration Class**

- Attributes: productID, designReference
- Methods: addToCart(), sendProductData()

3.8.3 Sequence Diagram

A Sequence Diagram illustrates how objects interact with each other in a specific sequence of operations. It focuses on the order in which messages are exchanged between system components.

In the TryVirtual system, the sequence of interactions typically follows these steps:

- The user opens the apparel customization page.
- The system loads the design editor interface.
- The user uploads images or adds text to the design canvas.
- The system processes the design data.
- The user requests a 3D preview of the customized apparel.
- The system maps the canvas design onto the 3D model.
- The rendered preview is displayed to the user.
- The user adds the customized product to the shopping cart.

3.8.4 Activity Diagram

An Activity Diagram represents the workflow of activities within the system. It describes the step-by-step process involved in completing a specific operation.

The activity flow of the TryVirtual system includes the following steps:

- User accesses the apparel customization page.
- System loads the design editor.
- User performs customization operations.
- System updates the design preview in real time.
- User selects the 3D preview option.

-
- System renders the customized design on the 3D model.
 - User reviews the design and confirms it.
 - Customized product is sent to the shopping cart.

3.9 User Interface Design

User Interface (UI) design is an important aspect of any web application because it directly affects user interaction and experience. The TryVirtual platform is designed with a modern and intuitive interface that allows users to customize apparel easily.

The user interface focuses on simplicity, clarity, and responsiveness so that users can perform customization tasks without difficulty.

The major design considerations for the user interface include:

- **Clean Layout:** The interface uses a structured layout with clearly separated sections for navigation, design tools, and preview.
- **Responsive Design:** The interface adapts to different screen sizes including desktops, tablets, and mobile devices.
- **Interactive Controls:** Buttons, sliders, and editing tools provide immediate visual feedback when users interact with them.
- **Customization Workspace:** The central area of the interface contains the design canvas where users can create and modify apparel designs.
- **Toolbars and Panels:** Sidebars and toolbars provide options for adding text, uploading images, and adjusting design elements.
- **Preview Options:** Users can view their design in both 2D and 3D preview modes.

The user interface is implemented using modern web technologies such as Next.js, Tailwind CSS, and reusable React components.

3.10 Database Design

Database design defines how data is structured and stored within the system. The TryVirtual platform requires data storage for managing user information, customization designs, and draft designs.

The database structure is designed to ensure efficient storage and retrieval of data while maintaining data consistency.

Key components of the database include:

- **User Table**

- UserID
- Username
- Email
- Password

- **Design Table**

- DesignID
- UserID
- CanvasData
- CreationDate

- **Draft Table**

- DraftID
- UserID
- DraftData
- LastModified

- **Product Table**

- ProductID
- ProductName
- ProductType
- ProductColor

The database ensures that user designs can be saved, retrieved, and processed efficiently.

3.11 Normalization

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity.

The database design of the TryVirtual system follows standard normalization techniques.

The normalization process includes the following stages:

- **First Normal Form (1NF)**

- Ensures that each table column contains atomic values.
- Removes repeating groups of data.

- **Second Normal Form (2NF)**

- Eliminates partial dependencies.
- Ensures that all non-key attributes depend on the primary key.

- **Third Normal Form (3NF)**

- Removes transitive dependencies.
- Ensures that all attributes depend only on the primary key.

Applying normalization improves database efficiency and ensures accurate data storage.

3.12 Database Manipulation

Database manipulation refers to the operations performed on the stored data within the system.

The TryVirtual platform performs several database operations including:

- Storing user customization designs.
- Saving draft designs for later editing.
- Retrieving saved designs when users reopen the application.
- Updating design data after modifications.
- Deleting unused or unwanted design drafts.

These operations ensure that user design data is managed efficiently throughout the customization process.

3.13 Methodology

The methodology used for developing the TryVirtual system focuses on structured development and iterative improvements.

The main steps followed during development include:

- **Requirement Analysis**

- Identifying the problem in existing apparel customization systems.
- Defining system requirements and objectives.

- **System Design**

- Designing system architecture and modules.
- Creating diagrams such as DFD, UML, and ER diagrams.

- **Implementation**

- Developing the web application using Next.js, TypeScript, and Tailwind CSS.
- Implementing the canvas-based design editor.
- Integrating the 3D visualization module using Three.js.

- **Testing**

- Verifying the functionality of each module.
- Testing user interactions and customization tools.
- Checking system performance and responsiveness.

- **Deployment**

- Deploying the application on cloud platforms.
- Ensuring accessibility and performance optimization.

The methodology ensures systematic development of the TryVirtual platform and allows continuous improvement throughout the project lifecycle.

Chapter 4

Implementation, Testing, and Maintenance

4.1 Introduction

The implementation phase of the software development life cycle involves converting the system design into a working software application. During this stage, developers write the code, integrate different modules, and ensure that the system performs according to the design specifications.

For the TryVirtual platform, the implementation focuses on developing a modern web-based application that enables users to customize apparel designs and visualize them in a realistic environment. The platform integrates technologies for user interface design, canvas-based editing, and three-dimensional rendering.

This chapter describes the tools, programming languages, and technologies used in the development process. It also explains the coding practices followed during development, the scheduling of project activities, and the testing techniques used to ensure the reliability of the system.

4.2 Languages, IDEs, Tools, and Technologies

The development of the TryVirtual platform required the use of several modern web technologies and development tools. These technologies were selected to ensure high performance, scalability, and ease of maintenance.

The major technologies used in the project are described below.

4.2.1 Programming Languages

- TypeScript

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- TypeScript is a strongly typed programming language built on top of JavaScript.
 - It improves code reliability and maintainability by introducing static type checking.
 - TypeScript is widely used in modern React and Next.js applications.

- **JavaScript**

- JavaScript is used for implementing dynamic functionality within the web application.
- It allows interaction with the browser and manipulation of the design canvas.

4.2.2 Frameworks and Libraries

- **Next.js**

- Next.js is a React-based framework used for building modern web applications.
- It supports server-side rendering and client-side rendering.
- It improves application performance and SEO optimization.

- **React**

- React is a JavaScript library used for building user interfaces.
- It allows developers to create reusable components.
- React improves the modularity and scalability of web applications.

- **Three.js**

- Three.js is a JavaScript library used for rendering 3D graphics in web browsers.
- It allows the system to display interactive 3D apparel models.
- It enables features such as model rotation, zooming, and texture mapping.

- **Fabric.js**

- Fabric.js is used for creating and manipulating objects on an HTML canvas.
- It allows users to add text, images, and shapes to the design area.
- It supports resizing, rotating, and positioning of objects.

4.2.3 Styling and UI Tools

- **Tailwind CSS**

- Tailwind CSS is a utility-first CSS framework.
- It allows rapid UI development using predefined classes.
- It helps maintain a consistent design throughout the application.

- **Lucide React Icons**

- Lucide icons provide lightweight and modern icon components.
- They enhance the visual appearance of the user interface.

4.2.4 Development Tools

- **Visual Studio Code**

- Visual Studio Code is the primary code editor used for development.
- It provides extensions for JavaScript, TypeScript, and React development.
- It offers debugging tools and integrated Git support.

- **Git and GitHub**

- Git is used for version control and tracking code changes.
- GitHub is used for hosting the project repository and managing collaboration.

- **Package Manager**

- A package manager is used to install and manage dependencies required by the project.
- It simplifies the process of adding libraries and frameworks to the application.

4.2.5 Deployment Platform

- **Vercel**

- Vercel is a cloud platform used for deploying modern web applications.
- It provides serverless deployment and automatic scaling.
- It integrates seamlessly with Next.js projects.

4.3 Coding Standards

Coding standards are a set of guidelines that developers follow while writing code. These standards help maintain consistency, readability, and maintainability within the project. Following coding standards also simplifies debugging and collaboration among developers.

The TryVirtual project follows several coding practices to ensure high-quality code.

- **Consistent Naming Conventions**

- Variables and functions use meaningful names.
- Component names follow PascalCase naming style.
- Variables and functions follow camelCase conventions.

- **Component-Based Structure**

- The application is divided into reusable components.
- Each component handles a specific functionality such as header, toolbar, or canvas editor.

- **Code Modularity**

- Logic is separated into smaller modules.
- Each module performs a specific task to improve maintainability.

- **Proper Documentation**

- Comments are included to explain complex sections of code.
- Important functions and modules are documented for future reference.

- **Error Handling**

- Error handling mechanisms are implemented to prevent system crashes.
- Proper messages are displayed to users in case of failures.

4.4 Project Scheduling

Project scheduling helps organize development tasks and ensures that the project is completed within the planned timeline. A structured schedule also helps track the progress of the project and manage development activities efficiently.

The TryVirtual project was developed in multiple stages, including planning, design, development, testing, and deployment.

The major phases of the project include:

-
- Requirement analysis and project planning.
 - System design and architecture preparation.
 - Development of the canvas-based customization editor.
 - Integration of the 3D visualization module.
 - Implementation of user interface components.
 - Testing and debugging of the system.
 - Deployment and documentation preparation.

4.5 Testing Techniques

Testing is an essential phase in software development that ensures the system works correctly and meets the required specifications. It helps identify errors, improve performance, and ensure reliability before deployment.

Different testing techniques were applied during the development of the TryVirtual platform.

- **Unit Testing**
 - Individual components of the system were tested separately.
 - Functions related to canvas editing and UI components were verified independently.
- **Integration Testing**
 - Multiple modules were tested together to ensure proper interaction.
 - The integration between the canvas editor and 3D preview module was verified.
- **System Testing**
 - The entire application was tested as a complete system.
 - The goal was to ensure that all functionalities work together correctly.
- **User Interface Testing**
 - The usability of the user interface was evaluated.
 - Layout responsiveness and user interactions were tested across devices.

- **Performance Testing**

- The system was tested for speed and responsiveness.
- Rendering performance of the 3D model was verified.

4.6 Test Plan

A test plan defines the strategy used for verifying the functionality of the system. It outlines the test cases and expected outcomes of the testing process.

The main objectives of the test plan for the TryVirtual platform include:

- Ensuring that the customization editor functions correctly.
- Verifying that users can upload images and add text elements to the canvas.
- Confirming that resizing and repositioning of design elements works properly.
- Checking that the 3D preview feature correctly displays the customized apparel.
- Verifying that the draft saving functionality stores designs successfully.
- Ensuring that the system performs correctly on different browsers and devices.
- Confirming that the customized product can be successfully sent to the shopping cart.

A successful test result indicates that the system meets the required specifications and provides a reliable user experience.

Chapter 5

Results and Discussions

5.1 Introduction

The results and discussion chapter presents the final outcome of the TryVirtual platform after successful implementation and testing. This chapter demonstrates how the developed system performs in real-world scenarios and how it addresses the problem identified in earlier chapters.

The TryVirtual platform successfully provides an interactive environment where users can customize apparel designs and visualize them before purchasing. The system integrates a canvas-based design editor along with a three-dimensional visualization module, enabling users to preview customized apparel in a realistic manner.

This chapter includes screenshots of the developed system, explanations of major modules, and observations about the system performance and user experience.

5.2 User Interface Representation with Snapshots

The user interface of the TryVirtual platform is designed to be simple, modern, and easy to use. The interface allows users to customize apparel designs using an interactive canvas editor and preview the final product.

The following screenshots represent the major screens of the system.

5.2.1 Homepage of the Customization Interface

The homepage of the customization platform displays the main workspace where users can begin designing their apparel. It includes navigation options, design tools, and a canvas area for editing.

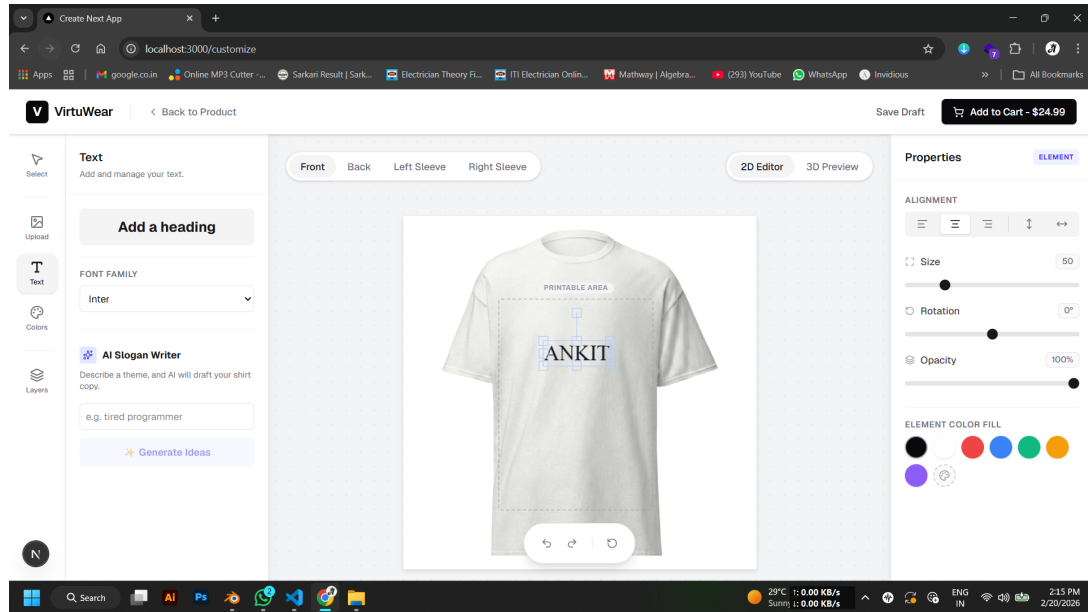


Figure 5.1: Homepage of the TryVirtual Customization Interface

5.2.2 Canvas-Based Design Editor

The canvas editor is the central workspace where users can add images, text, and design elements to the apparel template.

Key features of the canvas editor include:

- Uploading custom images for apparel printing.
- Adding text elements with customizable fonts and colors.
- Resizing and repositioning design objects.
- Rotating and scaling graphical elements.
- Real-time preview of the design.

5.2.3 Design Customization Tools

The platform provides a variety of customization tools that allow users to modify their design easily.

Some of the customization features include:

- Text editing tools for adding personalized messages.
- Image upload functionality for custom graphics.
- Color selection tools for adjusting design colors.

- Object transformation tools such as resize, rotate, and move.

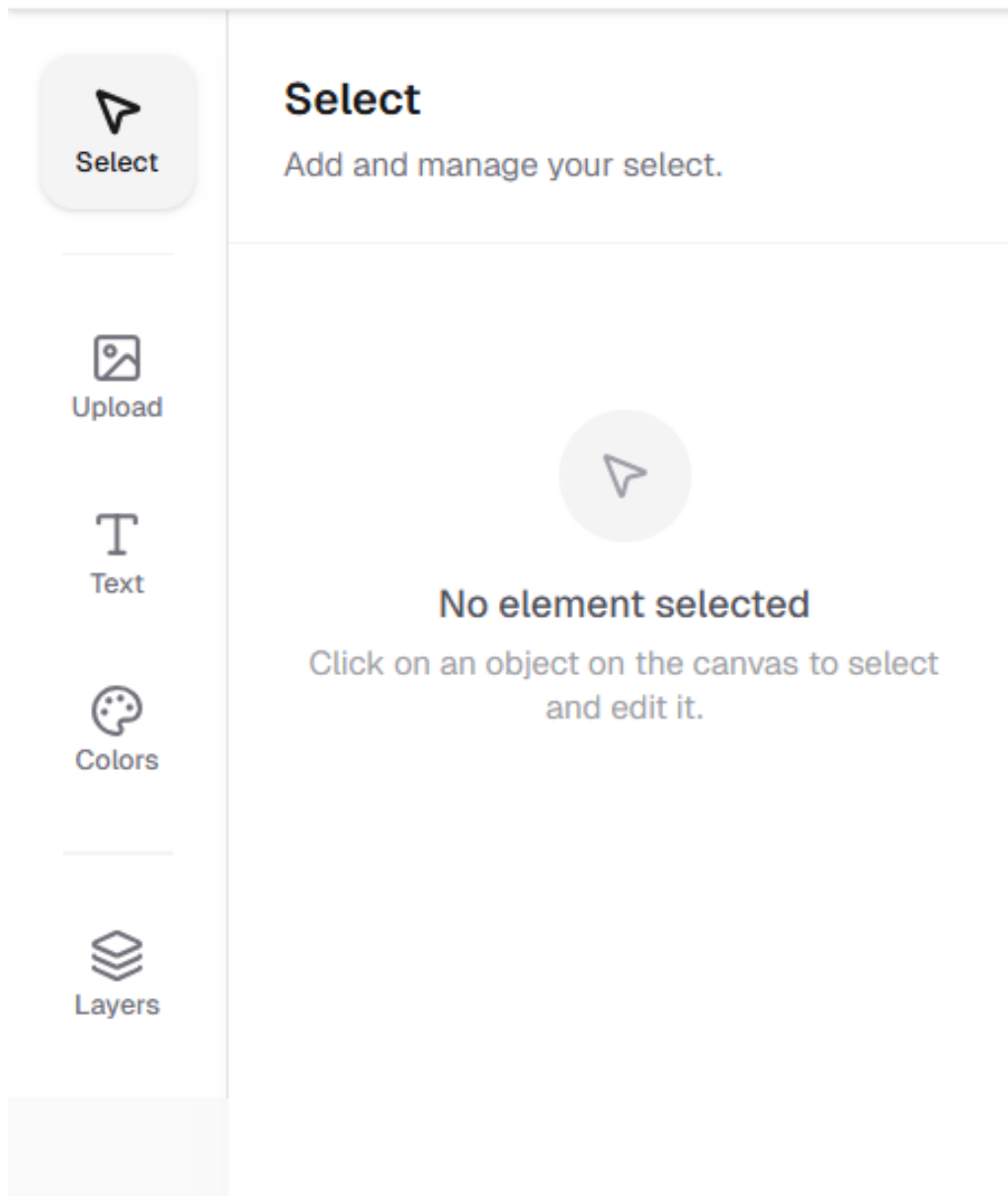


Figure 5.2: Customization Tools in TryVirtual Platform

5.2.4 3D Apparel Visualization

One of the key features of the TryVirtual platform is the ability to visualize the customized design on a three-dimensional apparel model.

The system converts the canvas design into a texture and applies it to the 3D model. Users can rotate the model and inspect the design from multiple angles.

Advantages of the 3D preview include:

- Realistic visualization of the final product.

-
- Ability to view the apparel from different perspectives.
 - Better understanding of how the design will appear on the actual garment.

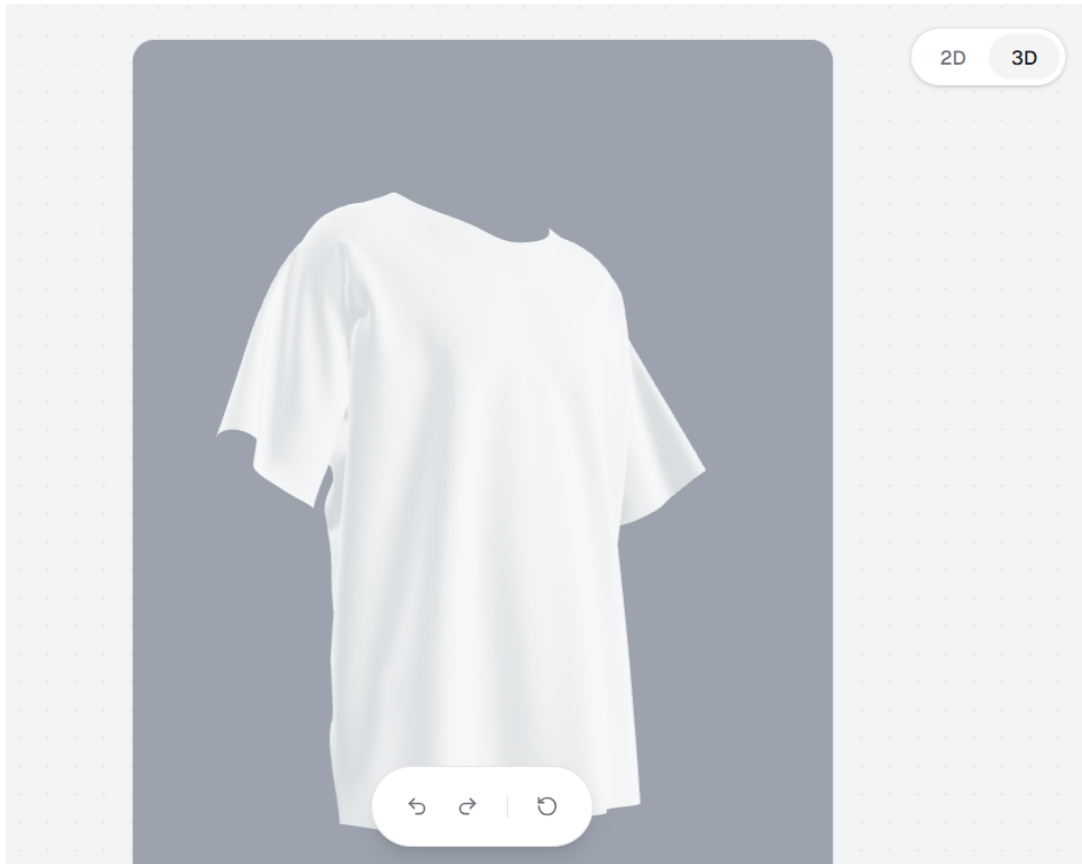


Figure 5.3: 3D Visualization of Customized Apparel

5.3 Description of Major Modules

The TryVirtual platform consists of multiple modules that work together to provide a seamless customization experience.

5.3.1 Customization Module

This module provides the design canvas and editing tools required for apparel customization.

Main functionalities include:

- Creating new apparel designs.
- Uploading images and adding text.
- Editing and arranging design elements.
- Managing object layers on the canvas.

5.3.2 Preview Module

The preview module allows users to view their customized apparel before finalizing the design.

Important features include:

- Real-time 2D preview of the design.
- Interactive 3D model rendering.
- Rotation and zoom controls for better visualization.

5.3.3 Draft Management Module

This module allows users to save their work and continue editing later.

Key functions include:

- Saving unfinished designs as drafts.
- Loading saved designs for further editing.
- Updating or deleting existing drafts.

5.3.4 E-Commerce Integration Module

This module connects the customization system with the existing e-commerce platform.

Its functions include:

- Sending customized product data to the shopping cart.
- Maintaining product configuration information.
- Supporting seamless integration with online stores.

5.4 Backend and Database Snapshots

The backend system manages the storage and retrieval of user customization data.

The database stores important information such as:

- User details.
- Saved design drafts.
- Canvas configuration data.
- Product customization information.

5.5 Discussion

The results of the TryVirtual platform demonstrate that the system successfully addresses the limitations of existing apparel customization platforms.

The major achievements of the system include:

- Providing a real-time apparel customization environment.
- Allowing users to design products using a canvas-based editor.
- Introducing 3D visualization for better product preview.
- Offering an interactive and responsive user interface.
- Supporting integration with existing e-commerce systems.

Overall, the developed system improves the online apparel customization experience and provides a more accurate representation of the final product before purchase.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

The TryVirtual platform was developed to address the limitations of existing apparel customization systems that only provide static two-dimensional previews of customized designs. Many online clothing platforms allow users to edit designs on a canvas but fail to provide an accurate visualization of how the final product will appear when printed on apparel.

The primary objective of this project was to develop an interactive web-based platform that allows users to customize apparel and visualize the final design using modern web technologies. The system successfully integrates a canvas-based customization editor with a three-dimensional visualization module, providing a more realistic preview of customized apparel.

The implementation of the TryVirtual platform demonstrates how modern web technologies can be used to improve the user experience in online product customization. By combining technologies such as Next.js, TypeScript, Tailwind CSS, Fabric.js, and Three.js, the system provides an efficient, responsive, and interactive environment for apparel design.

The major accomplishments of the project include:

- Development of a web-based apparel customization platform.
- Implementation of a canvas-based design editor that allows users to add images, text, and graphics.
- Integration of a 3D visualization module for realistic preview of customized apparel.
- Creation of a responsive user interface that works across multiple devices.
- Implementation of draft-saving functionality for storing incomplete designs.

-
- Integration capability with existing e-commerce systems for product purchasing.

The system successfully demonstrates how interactive design tools and three-dimensional visualization can enhance the online shopping experience by allowing users to better understand the final appearance of customized products.

6.2 Future Scope

Although the TryVirtual platform provides an effective solution for apparel customization and visualization, there are several opportunities for further improvements and enhancements in the future.

Some possible future developments include:

- **Augmented Reality (AR) Integration**

- Users could visualize apparel designs in real-world environments using mobile device cameras.
- This would allow customers to see how the apparel might look when worn.

- **Advanced 3D Clothing Simulation**

- Future versions could simulate fabric movement and realistic cloth behavior.
- This would improve the accuracy of apparel visualization.

- **AI-Based Design Assistance**

- Artificial intelligence could suggest design layouts, colors, and styles based on user preferences.
- AI tools could also automatically adjust design placements for better visual results.

- **User Account and Cloud Storage**

- Users could create accounts and store multiple designs securely in the cloud.
- This would allow them to access their designs from different devices.

- **Multi-Product Customization**

- The system could be expanded to support customization of additional products such as hoodies, caps, mugs, and phone cases.

- **Real-Time Collaboration**

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- Multiple users could collaborate on the same design project simultaneously.
 - This would be useful for teams or group projects.

By implementing these improvements, the TryVirtual platform could evolve into a powerful customization solution for the modern e-commerce industry.

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